Key Boundary & Control Classes: Boundary Classes

Class	Responsibility	Control
SearchHomeScreenUI	Captures destination, intended arrival time and duration; displays up to five ranked results and search warnings.	RecommendationController; PreferencesController
CarparkDetailViewUI	Shows carpark metadata, rates, current or predicted lots, confidence, timestamps, weighted explanation and actions.	RecommendationController; NavigationController; FavouritesController; ReportController
SettingsUI	Displays and edits vehicle, walking-distance, ranking-priority, reminder and saved-location settings.	PreferencesController
SessionUI	Displays the active or completed local parking session and its elapsed/accrued cost.	SessionController
FavouritesUI	Displays and removes device-local favourite carparks and labels.	FavouritesController
ReportSubmissionModalUI	Collects anonymous report category and description for the selected carpark.	ReportController
OperatorLoginUI	Collects the operator username and password.	AuthenticationController
OperatorConsoleUI	Displays report queue, override controls and per-source health panels.	AuthenticationController; OperatorAdminController; ReportController; APIHealthMonitor
CarparkDataInterface	Receives availability, reference and rate data from LTA DataMall, data.gov.sg and URA feeds.	ExternalDataController
MapServiceInterface	Resolves searched and saved destinations and supplies walking/driving route measures.	RecommendationController; PreferencesController
NavigationIntentInterface	Sends a verified carpark destination to the device's external navigation application.	NavigationController
NotificationInterface	Schedules and cancels the local parking reminder.	SessionController

Control Classes

Class	Responsibility	Use Case / Requirement
RecommendationController	Validates searches, resolves destinations, obtains evidence, filters candidates, predicts or falls back to availability, estimates cost and computes weighted rankings.	UC-01, UC-02
NavigationController	Launches external navigation; on unsupported-app or handoff failure, retains carpark details and provides the verified destination for manual use.	UC-03
PreferencesController	Validates and stores device-local preferences and saved locations; clears local data when requested.	FR-1.1 to FR-1.10
SessionController	Starts, extends and ends local parking sessions, calculates accrued cost and schedules reminders.	FR-6.1 to FR-6.11
FavouritesController	Adds and removes device-local favourites and labels.	FR-7.1 to FR-7.5
ReportController	Validates anonymous reports, enforces ReportQuota, queues reports, records operator resolutions and automatically rejects unmatched carpark identifiers.	UC-04, UC-06
AuthenticationController	Validates operator usernames/passwords over TLS and issues or revokes operator sessions.	UC-05
OperatorAdminController	Applies and withdraws closure/rate overrides and protects them from source refresh.	UC-06, UC-07
APIHealthMonitor	Collects and interprets per-source health, freshness and refresh outcomes for the operator panel.	UC-08
ExternalDataController	Polls external sources, normalizes payloads, records refresh outcomes and preserves active overrides.	UC-09